Normalizer fibres in rank-two Suzuki geometries

Human Compression project

Abstract

The families with a dihedral or a nonsplit torus normalizer as one parabolic can be recovered from two elementary constructions. A normalizer calculation gives uniqueness of the relevant subfield overgroups. The remaining configurations are fibres of an incidence relation between conjugacy classes of subgroups; these fibres are small quotients of a Suzuki root group. Their field automorphisms and inversion give the counts in the prime cases. This treats source Lemmas 6.2–6.6 and 6.8, with a correction to the prime-divisor condition in Table 6. The classical subgroup classification quoted as source Lemma 6.7 is retained as an input. The families with a subfield group as the only maximal parabolic, and the global classification, are not concluded here.

1 Scope and structural inputs

Write $q = 2^n$, where $n \geq 3$ is odd, and put

$$\theta = 2^{(n+1)/2}, \qquad a_\epsilon(q) = q + \epsilon\theta + 1 \quad (\epsilon \in \{+1, -1\}).$$

Let $G = \mathrm{Sz}(q)$. A configuration in this note is an unordered pair of proper subgroups $\{M, L\}$, with intersection $J = M \cap L$. Equivalence means the action of $\mathrm{Aut}(G)$, allowing interchange of the two subgroups. This is the specified-group convention of the source [**?**]; we do not identify geometries by arbitrary permutations which need not preserve their specified G action.

The following classical structure is retained, with its attribution to Suzuki, Tits and Lüneburg as in Sections 2.2 and 4 of the source. The nonidentity elements of G are partitioned by its Sylow two-subgroups and its cyclic tori of orders $q - 1, a_+(q), a_-(q)$. The normalizers of these parts are respectively

$$B = P \rtimes \mathbb{F}_q^*, \quad D = C_{q-1} \rtimes C_2, \quad N_\epsilon = C_{a_\epsilon(q)} \rtimes C_4. \tag{1}$$

These groups are maximal in G. Each indicated type of part forms one conjugacy class. Every subgroup of G either lies in one of these normalizers or is a conjugate of a standard subfield group $S = \mathrm{Sz}(s)$, where $s = 2^d \geq 8$ and $d \mid n$. In the latter case the conjugacy class for each s is unique. The same description holds inside S; its maximal proper simple subfield groups have prime field index. This is precisely the classical subgroup input quoted from Suzuki [**?**] as source Lemma 6.7, together with the maximal-subgroup list in Section 4. It is not proved anew here.

We also retain the standard action on the Suzuki ovoid: B fixes one point, a nonidentity involution fixes just one point, and a split torus fixes two points whose stabilizers are interchanged by D. The centralizer of an involution is its Sylow two-subgroup. Finally $\mathrm{Aut}(G) = G \rtimes \langle\phi\rangle$, where ϕ is coordinate Frobenius of order n.

We use the previously reviewed preliminary note for the pairwise coprimality of the three odd orders, their proper-subfield arithmetic, and the maximal subgroups of the solvable groups in (1). Explicitly, in $C_t \rtimes C_4$ these maximal subgroups are $C_t \rtimes C_2$ and $C_{t/p} \rtimes C_4$, with $p \mid t$ prime; in D they are C_{q-1} and $D_{2(q-1)/p}$. In a point stabilizer they are $P \rtimes C_{(q-1)/p}$ and $Z \rtimes C_{q-1}$, where $Z = Z(P)$. We write D_{2m} for a dihedral group of order $2m$. The arithmetic

also shows that if any one of $q-1, a_+(q), a_-(q)$ is prime, then n is prime: if n were composite, a proper subfield of order at least eight would make all three composite.

For rank two, the elementary coset criterion we use is as follows. If M is maximal in G, $L \not\leq M$, and $J = M \cap L$ is maximal in both M and L, then $\Gamma(G; M, L)$ is firm, connected, flag transitive and residually weakly primitive. Indeed, the local degrees are $[M : J]$ and $[L : J]$, both at least two; rank-two coset geometries are flag transitive; $\langle M, L\rangle = G$ gives connectivity; and the three required primitive coset actions are those with stabilizers $M < G$, $J < M$, and $J < L$. Conversely, these maximality conditions on J are necessary when M is a maximal parabolic.

2 Two root-group quotients

Use the standard root coordinates

$$P = \{u(a, b) : a, b \in F\}, \qquad F = \mathbb{F}_q, \qquad u(a, b)u(c, e) = u(a + c, b + e + ac^\theta). \tag{2}$$

Put $z = u(0, 1)$, $w = u(1, 0)$, $J_2 = \langle z\rangle$ and $J_4 = \langle w\rangle$. Then $w^2 = z$ and both J_2, J_4 are fixed pointwise by ϕ. We retain the classical realization of (2) in the faithful Suzuki matrices, rather than reconstructing the ambient simple group.

Lemma 1. *The normalizers of these two subgroups are*

$$\mathrm{N}_G(J_2) = P, \qquad \mathrm{N}_G(J_4) = R := \{u(a, b) : a \in \mathbb{F}_2,\ b \in F\}. \tag{3}$$

The group R is abelian of order $2q$ and is isomorphic to $C_4 \times C_2^{n-1}$. The quotient R/J_4 is naturally the additive group $F/\mathbb{F}_2$. The cyclic subgroups of order four containing J_2 are parametrized by $F/\mathbb{F}_2$, the four-groups containing J_2 by $(F/\mathbb{F}_2) \setminus \{0\}$, and the subgroups of order eight containing J_4 by the same nonzero quotient. These parametrizations commute with Frobenius. Also

$$P^\phi = R^\phi = J_4. \tag{4}$$

Proof. The first equality in (3) is the retained involution-centralizer fact, since an automorphism of a group of order two is trivial. Every element normalizing J_4 fixes $w^2 = z$, so lies in P. The multiplication law gives

$$u(a, b)^2 = u(0, a^{\theta+1}), \qquad [u(a, b), w] = u(0, a + a^\theta), \tag{5}$$

where the choice of commutator convention does not change the second formula because its value has order at most two. Conjugation preserves the first coordinate of w, so normalizing J_4 is equivalent to $a + a^\theta \in \mathbb{F}_2$. The absolute trace of $a + a^\theta$ is zero, whereas the trace of 1 is one because n is odd. Thus $a = a^\theta$. The fixed field of $x \mapsto x^\theta$ is $\mathbb{F}_2$, since $\gcd(n, (n+1)/2) = 1$. This proves (3). Formula (2) on first coordinates in $\mathbb{F}_2$ proves that R is abelian, has exponent four, and has squares exactly J_2. Its stated abstract structure and $R/J_4 \cong F/\mathbb{F}_2$ follow, the latter map being represented by $u(0, b) \mapsto b + \mathbb{F}_2$.

The exponent $\theta + 1$ is relatively prime to $q - 1$: a common odd divisor would give $\theta = -1$ and $\theta^2 = 2$ modulo that divisor, hence $1 = 2$. Thus (5) shows that an element whose square is z has first coordinate one. The groups $\langle u(1, b)\rangle$ are therefore all cyclic groups of order four containing J_2, and two values of b give the same group precisely when they differ by one. All involutions belong to $Z = \{u(0, b) : b \in F\}$, so four-groups containing J_2 are the two-dimensional binary subspaces of F containing one. These are parametrized by the nonzero elements of $F/\mathbb{F}_2$.

If a group of order eight contains J_4, that index-two subgroup is normal; the group is therefore in R. Conversely, inverse images of order-two subgroups of R/J_4 have order eight. They are $C_4 \times C_2$ since R is abelian of exponent four. All formulas commute with squaring of field coordinates. Formula (4) follows because the only fixed coordinates are zero and one. □

Lemma 2. *Suppose n is prime. Frobenius has the following numbers of fixed cosets:*

$$\begin{array}{c|ccc} X & P/J_2 & P/J_4 & R/J_4 \\ \hline |X| & q^2/2 & q^2/4 & q/2 \\ |X^\phi| & 2 & 1 & 1. \end{array} \tag{6}$$

On $(F/\mathbb{F}_2) \setminus \{0\}$ it has no fixed point. On $(P/J_2) \setminus \{1\}$ modulo inversion, it has exactly one fixed orbit. The number of inversion orbits in this last set is

$$q^2/4 + q/2 - 1. \tag{7}$$

Proof. For either pointwise fixed subgroup $J = J_2, J_4$, a fixed right coset gJ gives $\phi(g) = gh$ for some $h \in J$. Iterating n times gives $h^n = 1$, so $h = 1$ because n is odd and h has two-power order. Hence fixed cosets are exactly the cosets represented by fixed elements. Equation (4) proves (6). The statement for $F/\mathbb{F}_2$ follows from R/J_4.

Let $H = P/J_2$. It has q elements with square one: by (5), there are $2q$ elements of P with square in J_2, with first coordinate zero or one, and each coset has two elements. Inversion on the set H fixes exactly these q elements. Pairing the others and removing the identity gives $(|H| + q)/2 - 1$, which is (7). An inversion orbit fixed by ϕ has size at most two. The permutation of it induced by an element of odd prime order cannot exchange its two members; each member is therefore fixed. By (6), H^ϕ has order two, and its sole nonidentity element gives the one fixed orbit. □

3 Uniqueness of three subfield overgroups

Lemma 3. *Let a finite group K have one conjugacy class of subgroups S of the specified type. Suppose $H < S$, and the subgroups of S of the specified H type form one S-class. Then the number of conjugates of S containing a fixed conjugate of H is*

$$\frac{|S|\,|\,\mathrm{N}_K(H)|}{|\,\mathrm{N}_K(S)|\,|\,\mathrm{N}_S(H)|}. \tag{8}$$

In particular it is one if S and H are self-normalizing in K.

Proof. Count incident pairs of conjugates (S', H') with $H' \leq S'$. There are $|K|/|\,\mathrm{N}_K(S)|$ first entries, each incident with $|S|/|\,\mathrm{N}_S(H)|$ second entries. Divide by the total number $|K|/|\,\mathrm{N}_K(H)|$ of second entries. □

Theorem 4. *Let $K = \mathbb{F}_s \subseteq F$, with $s \geq 8$. Each subgroup of one of the following types is contained in exactly one subgroup isomorphic to* $\mathrm{Sz}(s)$*:*

$$B_s = U(K, K) \rtimes K^*, \qquad D_{2(s-1)}, \qquad C_{a_\delta(s)} \rtimes C_4 \quad (\delta = +1, -1). \tag{9}$$

Proof. All the groups in (9) occur in the standard $S = \mathrm{Sz}(s)$. Their respective conjugacy classes in S are unique by the retained structure. We prove that each is self-normalizing in G. The subfield group S is also self-normalizing. Indeed, its normalizer contains the nonsolvable simple group S, so cannot be contained in a solvable normalizer in (1). The subgroup classification makes it a simple subfield group; since it normalizes its nontrivial normal subgroup S, it equals S.

For B_s, its characteristic Sylow two-subgroup determines the unique fixed ovoid point, so any normalizer lies in $B = PH$. Write its elements using $u(a, b)$ and $h(t)$, where

$$u(a, b)^{h(t)} = u(t^\theta a, t^{\theta+2} b). \tag{10}$$

The exponent $\theta+2$ is coprime to $q-1$: a common odd divisor would give $\theta=-2$ and $\theta^2=2$, hence $4=2$. Preserving the centre $U(0,K)$ of the Sylow subgroup in B_s therefore forces $t^{\theta+2}\in K^*$, and hence $t\in K^*$. Remove this factor. For $k\in K^*\setminus\{1\}$, the first coordinate of the commutator of $u(a,b)$ with $h(k)$ is a nonzero K-multiple of a. Membership in B_s forces $a\in K$. Remove the factor $u(a,0)\in B_s$ as well. The remaining central factor $u(0,b)$ has commutator coordinate $(k^{\theta+2}-1)b$, up to multiplication by a nonzero element of K. This forces $b\in K$. Thus $\mathrm{N}_G(B_s)=B_s$.

For either odd normalizer in (9), its cyclic odd subgroup is characteristic. The partition into tori forces its normalizer to preserve the unique ambient torus containing that subgroup. In the split case this gives a subgroup of $C_{q-1}\rtimes C_2$. A torus element x normalizes $D_{2(s-1)}$ precisely when $x^2\in C_{s-1}$; since the torus has odd order this is equivalent to $x\in C_{s-1}$.

In the nonsplit case the ambient torus normalizer is $C_t\rtimes\langle v\rangle$, with $t=a_\epsilon(q)$ and $|v|=4$. The standard action on the cyclic torus is the power q or its inverse. Both $q-1$ and $q^{-1}-1$ are coprime to t, because $\gcd(q-1,t)=1$. For $m=a_\delta(s)$, a torus element x normalizes $C_m\rtimes\langle v\rangle$ precisely when the commutator of x with v lies in C_m. This is equivalent to $x\in C_m$. All complements of order four are torus conjugate, so this calculation applies to the subgroup in (9). It is self-normalizing. Equation (8) now proves all three uniqueness assertions. □

These are source Lemmas 6.5, 6.6 and 6.8, including the version of 6.8 for the other nonsplit torus. The common step is incidence counting after self-normalization. No new conjugacy classification of subfield groups is being asserted.

4 Fibres of subgroup incidence

Fix $J=J_2$ or J_4. If the copies of J in a subgroup M form one conjugacy class in M, then the conjugates of M containing this fixed J form one $\mathrm{N}_G(J)$-orbit. To prove this, first conjugate one copy of M to another; conjugate again within the second copy to send the image of J to J. The stabilizer is $\mathrm{N}_G(J)\cap\mathrm{N}_G(M)$. Thus the orbit is an actual coset space, not merely a set of the same cardinality.

Applied to (1) and to its prime subgroups, this gives

J	M	$\mathrm{N}_G(J)\cap\mathrm{N}_G(M)$	fibre
J_2	$D_{2(q-1)}$	J_2	P/J_2
J_2	$D_{2p},\ p\mid a_\epsilon(q)$ prime	C_4	P/C_4
J_4	$C_t\rtimes C_4$	J_4	R/J_4.

(11)

In the last row t may be the whole nonsplit torus order or a prime divisor of it. For the second row, the full normalizer of D_{2p} is $C_p\rtimes C_4$. This follows by preserving its characteristic odd subgroup, then using the odd-order squaring argument from Theorem **??**. Its stabilizer of its given involution is a cyclic group of order four. For the last row, the commutator argument in the same theorem shows that $C_t\rtimes C_4$ is self-normalizing, even when t is a proper prime divisor of the ambient torus order. The required conjugacy of involutions or complements inside these subgroups follows from the cyclic semidirect products of odd order by C_2 or C_4. This also follows from the affine complement argument in the preliminary note.

Suppose now that n is prime. Every fibre in (11) has a Frobenius-fixed member: its cardinality is respectively $q^2/2$, $q^2/4$ or $q/2$, congruent to $2,1,1$ modulo n by Fermat's congruence. The orbits of a cyclic group of prime order have size one or n, so a fixed member exists. Choose that member as base. In the second row its stabilizer C_4 is Frobenius invariant, and Frobenius acts trivially on it because $\mathrm{Aut}(C_4)$ has order two. By (4), this stabilizer is the particular J_4 used above. Thus (11) is Frobenius equivariant with exactly the quotients in (6); no assumption about a preferred complement is needed.

For two distinct dihedral parabolics with common J_2, the first fibre in (11) is a regular set for $H = P/J_2$. Simultaneous translation sends an ordered pair to $(1, h)$ with $h \neq 1$; interchanging its entries sends h to h^{-1}. Hence the unordered configurations are the inversion classes in $H \setminus \{1\}$, followed by Frobenius. For two equal types of nonsplit normalizer, the same argument uses R/J_4, an elementary abelian group, so inversion does nothing.

For distinct parabolic types there is no interchange. Once the first parabolic containing J_2 is fixed, its stabilizer in P is J_2, which acts trivially on P/J_4 and on the order-four groups in Lemma **??**. The nonsplit calculation has the analogous trivial action of J_4 on R/J_4. These observations explain why the small fibres give the complete configuration counts, rather than just numbers of possible second subgroups.

5 The split normalizer

Theorem 5. *Take $M = D_{2(q-1)}$. If $q-1$ is prime, the possible second parabolics and their numbers of equivalence classes are precisely the rows of the following table. Each prime p in the third row gives a separate family.*

L	J	*number*
$Z \rtimes C_{q-1}$	C_{q-1}	1
$D_{2(q-1)}$	C_2	$1 + \dfrac{(q-2)(q+4)}{4n}$
$D_{2p},\ p \mid a_+(q)a_-(q)$	C_2	$1 + \dfrac{(q-2)(q+2)}{4n}$
C_2^2	C_2	$\dfrac{q-2}{2n}$
C_4	C_2	$1 + \dfrac{q-2}{2n}.$

(12)

If $q-1$ is composite, the first row still occurs once. The only other possibilities are

$$L = \mathrm{Sz}(s), \qquad J = D_{2(s-1)}, \qquad (q-1)/(s-1) \text{ prime}, \tag{13}$$

with $s \geq 8$ a proper subfield order. Each occurs once.

Proof. First consider $J = C_{q-1} = A$, which is always maximal in M. A proper overgroup L of A not in M must lie in a point stabilizer, by the subgroup classification and coprimality of the three odd factors. A proper subfield group cannot contain A: if $q \geq s^3$, every cyclic torus in $\mathrm{Sz}(s)$ has order less than $q-1$. The torus A fixes two ovoid points, giving the two possible point stabilizers.

In either one write $L = VA$, with $V = L \cap P$ invariant under the full scalar torus. If $V \leq Z$, nonzero invariance forces $V = Z$, since the action is all nonzero scalars. If $V \not\leq Z$, the same argument on P/Z shows $VZ = P$. The Frattini property $Z = \Phi(P)$ from the preliminary note then gives $V = P$. But $A < ZA < PA$, so A would not be maximal in L. Consequently $L = ZA$. The two choices are interchanged by M, and their intersection with M is exactly A. This proves the first row for every q.

Suppose $q-1$ is prime. The other maximal intersection in M is J_2. We classify proper overgroups L in which J_2 is maximal. If L is contained in a point stabilizer, its normal two-subgroup $L \cap P$ contains J_2. By maximality it either equals J_2 or equals L. In the first case L normalizes J_2, hence lies in its two-group centralizer P as well. In both cases L is a two-group, and maximality makes its order four. It is therefore C_4 or C_2^2. The cyclic normalizers give D_{2p}, with p prime, or C_4. The split case has $p = q-1$; the other two odd orders give the third row. No simple subfield group has a maximal subgroup of order two, by its retained maximal-subgroup list. This proves exhaustion. The intersections in all distinct-type cases

are exactly J_2 by their orders. Two distinct split normalizers containing J_2 cannot have a common odd element, since such an element lies in a unique torus; their intersection is also exactly J_2. All the groups are outside M, so the coset criterion applies.

Here n is prime. The inversion quotient for the second row has size (7) and one fixed point by Lemma **??**. An action of order n on a finite set X has $(|X| + (n-1)|X^\phi|)/n$ orbits. This gives $1 + (q-2)(q+4)/(4n)$. The third row uses P/J_4, of size $q^2/4$ with one fixed point, giving $1 + (q-2)(q+2)/(4n)$. The last two rows use respectively the nonzero quotient $F/\mathbb{F}_2$ and the whole quotient, of size $q/2 - 1$ and $q/2$, with zero and one fixed points. This proves all counts in (12).

Finally suppose $q - 1$ is composite. Apart from A, a maximal intersection has form $J = D_{2m}$, with $m = (q-1)/p > 1$ and p prime. Any overgroup of J lying in a solvable normalizer must lie in M: the characteristic odd rotation subgroup determines its ambient split torus; a point stabilizer cannot contain its inverting involution, since its quotient by the normal two-group is abelian of odd order. Coprimality excludes nonsplit tori. Thus an overgroup outside M is $L = \mathrm{Sz}(s)$. Maximality of J in L forces $m = s - 1$ by its classical list. Conversely the standard such J occurs when the condition in (13) holds. It lies in a unique L by Theorem **??**. Its conjugacy class in M is unique, so this produces exactly one equivalence class. The maximality conditions and exact intersection follow as before. □

Theorem **??** recovers source Lemma 6.2 and Tables 2 and 3. The common proof does not separately enumerate the subgroups of each ambient Suzuki group.

6 The nonsplit normalizers

Theorem 6. *Let $t = a_\epsilon(q)$ and $M = C_t \rtimes C_4$. If t is prime, the complete list is*

L	J	*number*
$C_t \rtimes C_4$	C_4	$\dfrac{q-2}{2n}$
$C_p \rtimes C_4,\ p \mid a_{-\epsilon}(q)$ *prime*	C_4	$1 + \dfrac{q-2}{2n}$
$C_4 \times C_2$	C_4	$\dfrac{q-2}{2n}$.

(14)

If t is composite, the only possibilities are

$$L = \mathrm{Sz}(s), \qquad J = C_{a_\delta(s)} \rtimes C_4, \qquad t/a_\delta(s) \text{ prime}, \tag{15}$$

where $s \geq 8$ is a proper subfield order and $a_\delta(s) \mid t$. Each has one equivalence class.

Proof. One maximal subgroup of M is $C_t \rtimes C_2$. Its nontrivial characteristic odd subgroup determines the ambient nonsplit torus. Any overgroup outside M must therefore be a simple subfield group, by the same partition and coprimality argument as above. For $C_t \rtimes C_2$ to be maximal in $\mathrm{Sz}(s)$ the classical list would require $t = s - 1$. This is impossible: $s - 1 \mid q - 1$, whereas $\gcd(t, q-1) = 1$ and $t > 1$.

If t is prime, the remaining maximal intersection is J_4. A proper overgroup L in which J_4 is maximal, if contained in a point stabilizer, must be a two-group. Indeed its normal two-subgroup contains J_4, and if it equals J_4 the whole group lies in $\mathrm{N}_G(J_4) = R$, still a two-group. Maximality makes $|L| = 8$, whence Lemma **??** gives $C_4 \times C_2$. A split torus normalizer has no element of order four. A nonsplit one gives $C_p \rtimes C_4$, with p prime. The prime $p = t$ gives another conjugate of M; all other primes must divide the other nonsplit torus order. The classical maximal list excludes a simple subfield group with maximal subgroup C_4. This proves exhaustion. Distinct conjugates of M containing J_4 intersect exactly in J_4, since any

common odd element would determine the same torus and normalizer. Different types have intersection J_4 by their orders.

Since n is prime, the first row is the nonzero elementary abelian quotient R/J_4 modulo Frobenius, giving $(q-2)/(2n)$. The second is the whole quotient, with one fixed point, giving $1+(q-2)/(2n)$. The third is the nonzero quotient from Lemma **??**. These are actual incidence fibres by (11), and the interchange argument has already been included.

If t is composite, the remaining maximal intersections are $J = C_m \rtimes C_4$, where $m = t/p > 1$. Their nontrivial odd subgroups again exclude any solvable overgroup outside M. Thus $L = \mathrm{Sz}(s)$, and maximality in its classical list forces $m = a_\delta(s)$. Conversely each condition (15) gives the standard subgroup and its unique subfield overgroup by Theorem **??**. Conjugacy within M and this uniqueness give one class. The coset criterion proves the claimed geometric properties in every listed case. □

For $\epsilon = +1$, this gives source Lemma 6.3 and Tables 4–5. For $\epsilon = -1$, it gives Lemma 6.4 and Tables 6–7, subject to the following explicit correction. The second row of printed Table 6, page 110, requires its prime p to divide β_q, the source's notation for $a_-(q)$. That table already assumes β_q prime, so this repeats the same-normalizer family in its first row and omits the opposite torus family. The condition must be $p \mid \alpha_q = a_+(q)$, as (14) proves. At $q = 8$ the missing family has parabolics $C_5 \rtimes C_4$ and $C_{13} \rtimes C_4$. It is not a new conjecture or a global classification correction; it is a corrected condition in this particular table.

7 Attribution and remaining obligations

The recovered assertions are source Lemmas 6.2–6.6 and 6.8, and Tables 2–7, with the Table 6 condition just specified. The source's Lemma 6.7 remains an explicitly attributed classical input from Suzuki. Our contribution in this component is the common derivation through normalizer incidence, the two root-group quotients and their Frobenius actions. We make no priority claim for incidence counting, Burnside's orbit formula, or these elementary group-theoretic tools. The source's original case proofs and numerical assertions remain attributed to [**?**].

No part of this note supplies the remaining global classification. In particular, configurations whose only maximal parabolic is a subfield group require separate treatment of all its maximal intersections. The already reviewed scalar-subspace correction addresses one such family and shows that the printed global classification cannot be retained unchanged. Neither the present note nor that correction alone settles all remaining cases or the source's global numerical tables. Finite controls of the quotient formulas, when attached, are checks of the written proof; they do not replace its quantified arguments. This component has no Lean certificate and carries no whole-original completion credit by itself.

References

[1] D. Leemans, *The rank 2 geometries of the simple Suzuki groups* $\mathrm{Sz}(q)$, Beiträge zur Algebra und Geometrie 39 (1998), no. 1, 97–120.

[2] M. Suzuki, *On a class of doubly transitive groups*, Annals of Mathematics 75 (1962), 105–145.